

TILAK RAGHUNANDAN G

Software Engineer • AI Platform Engineer • GenAI & LLM Systems • Full-Stack Developer

tilakraghunandan0106@gmail.com | +91 8317338824 | Shivamogga, Karnataka, India | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

INTRODUCTION

Software Engineering undergraduate with hands-on experience building AI/GenAI systems — RAG pipelines, LLM integration, and prompt engineering — alongside full-stack development with React.js, Node.js, and REST APIs. Strong foundations in Data Structures, OOP, and databases, backed by 3 podium finishes at state- and national-level hackathons and production-grade projects reaching 95% model accuracy. Eager to contribute as a Software Engineer or AI Platform Engineer, building scalable, AI-assisted technology solutions.

EDUCATION

B.E. — Artificial Intelligence & Machine Learning Jan 2023 – Jan 2027 (Expected)

PES Institute of Technology and Management, Shivamogga

CGPA: 7.35 / 10 • Coursework: AI, ML, Deep Learning, Computer Vision, DSA, DBMS, OS, Linear Algebra & Statistics

Pre-University Education (PCMCs / Science) Jan 2022 – Jan 2023

DVS Composite PU College, Shivamogga

SKILLS

- **Programming Languages:** Python (Primary), Java (Basic), C, JavaScript
- **AI, GenAI & LLMs:** Prompt Engineering, RAG Pipelines, LLM Integration, AI Agents, Embeddings, Semantic Search, Vector Databases, Hugging Face
- **Software Engineering & Full-Stack:** Data Structures & Algorithms, OOP, React.js, Node.js, REST APIs, HTML5/CSS3, PHP
- **Databases & Cloud:** MySQL, MongoDB, Vector DB, Relational Schema Design, Google Cloud (GCP)
- **Computer Vision / ML:** PyTorch, Scikit-learn, CNNs, Transfer Learning, OpenCV
- **Tools:** Git/GitHub, VS Code, Postman, Linux CLI, n8n

PROJECTS

AI Retinal Disease Detection System 2025

Python • PyTorch • CNN • ResNet • VGG16 • OpenCV

- Built a multi-class retinal disease classifier (transfer learning + U-Net segmentation) achieving 95% accuracy with full ROC-AUC / F1 evaluation.

Multilingual Emergency Support Platform 2025

Top 5 Finalist, Bapuji Institute of Technology

React.js • Node.js • NLP • LLM APIs • Speech-to-Text

- Built a full-stack rural healthcare platform with real-time speech-to-text, multilingual NLP translation, and LLM-based medical triage via REST APIs.

EV Price Prediction Bot 2025

Python • Scikit-learn • Pandas • Regression Modeling • GitHub

- Built a regression-based ML pipeline to predict electric vehicle prices from technical specifications and market data, with feature engineering and model evaluation for accurate estimation.

ACHIEVEMENTS

- 1st Runner-Up — Presidency University Hackathon (48-hr sprint), beat 100+ teams (Mar 2026)
- 2nd Runner-Up — DIGITISE 2K25, real-time CV pipeline for agri quality control, 80+ teams (Oct 2025)
- Top 5 Finalist — Bapuji Institute of Technology Hackathon, Multilingual Emergency Support Platform (May 2025)
- National-Level Participant — MONAITHON 2025 (JNNCE) • Agentic AI Day (Google Cloud × Hack2Skill) • GDG Solution Challenge 2025 (2025)

EXPERIENCE

AI / ML Engineering Intern Oct 2025 – Feb 2026

Unriddle Technologies, Shivamogga

- Designed and deployed end-to-end RAG pipelines using LLMs, vector embeddings, and semantic search over large document corpora, cutting retrieval latency and improving answer relevance.
- Applied prompt engineering and fine-tuned retrieval parameters to boost Q&A accuracy; built NLP preprocessing/chunking workflows in an Agile team environment.

Volunteer Coordinator Sep 2024 – Present

Anveshana Startup Summit, PES Institute of Technology and Management